Pergamon

Scripta Metallurgica *et* Materialia, Vol. 33, No. 7, pp. 1117-1125, 1995
Elsevier Science Ltd
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0956-716X/95 $9.50 + .00

0956-716X(95)00327-4

OXIDATION BEHAVIOR OF TiAl-X (X=Cr, V, Si, Mo or Nb) INTERMETALLICS AT ELEVATED TEMPERATURE

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(Received January 3, 1995)
(Revised May 19, 1995)

Abstract

Oxidation behaviour of TiAl intermetallic compounds with the additions of Cr, V, Si, Mo, or Nb was investigated at 900~1100°C under the atmospheric environment. The reaction products were examined by XRD and SEM equipped with WDX. The weight gain by continuous oxidation increased with the addition of Cr or V, but there was less weight gain when Mo, Si or Nb was added individually. Thus, it is concluded that the addition of Cr or V did not improve the oxidation resistance, whereas the addition of Si, Mo or Nb improved it. Nb strengthened the tendency to form Al_2O_3 in the early stage of oxidation, due to the formation of the continuous Al_2O_3 and dense TiO_2+Al_2O_3 layers. According to the Pt-marker test of TiAl-5wt%Nb, oxides were formed mainly by inward diffusion of oxygen. Upon thermal cyclic oxidation at 900°C, it was shown that the addition of Cr or Nb improved the adherence of oxide scale to the substrate.

Introduction

Titanium-aluminide intermetallics have an excellent potential to become one of the most important aerospace materials because of their low density, high melting temperature, good elevated temperature strength and modulus retention, high resistance to oxidation and hydrogen absorption, and excellent creep properties (1). Titanium with a low aluminum content and Ti_3Al alloy are already widely used because of excellent mechanical properties, but are limited to low temperature applications due to insufficient oxidation resistance. Titanium alloys with higher aluminum content, such as TiAl and $TiAl_3$ compounds, are therefore being developed for high temperature uses (2-15). However, in the case of TiAl, poor oxidation resistance currently limits the use of TiAl above about 800°C because the external layer formed on these compounds is not a

TABLE 1
Chemical Compositions of TiAl Alloys Analyzed by EMPA (wt%)

	Ti	Al	Cr	V	Si	Mo	Nb
TiAl	60.7	39.3	—	—	—	—	—
10wt%Cr	56.48	33.76	9.76	—	—	—	—
5wt%Cr	56.02	39.52	4.46	—	—	—	—
5wt%V	58.72	36.76	—	4.79	—	—	—
5wt%Si	58.72	36.76	—	—	4.52	—	—
5wt%Mo	59.41	35.62	—	—	—	4.52	—
5wt%Nb	59.41	36.03	—	—	—	—	6.09

protective Al_2O_3 scale, but also TiO_2 or mixture of Al_2O_3 and TiO_2. Some methods have been proposed for improving the oxidation resistance of TiAl. One of those is the addition of the 3rd element to TiAl to enhance the formation of a protective Al_2O_3 layer. The oxidation behaviour of TiAl intermetallics was known to be affected significantly by the addition of ternary elements(16-20). The 3rd element addition, such as W, Nb and Mo, was known to improve oxidation resistance of TiAl, but to reduce room temperature ductility. On the other hand, V, Cr and Mn could improve room temperature ductility, but was known to decrease oxidation resistance of TiAl.

The present study is directed toward studying the effect of the addition of 3rd elements, such as Cr, V, Si, Mo and Nb, on the isothermal and thermal cyclic oxidation behaviour of TiAl intermetallic, in order to improve the oxidation resistance of TiAl at elevated temperature.

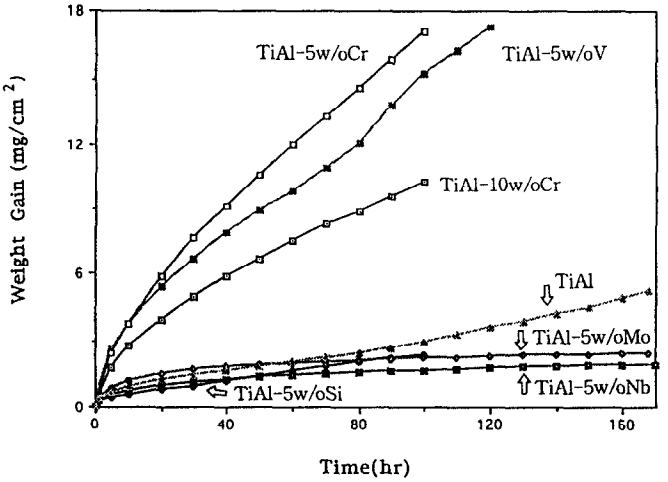


Figure 1. Weight change per unit area vs. oxidation time for samples oxidized in air at 900°C.

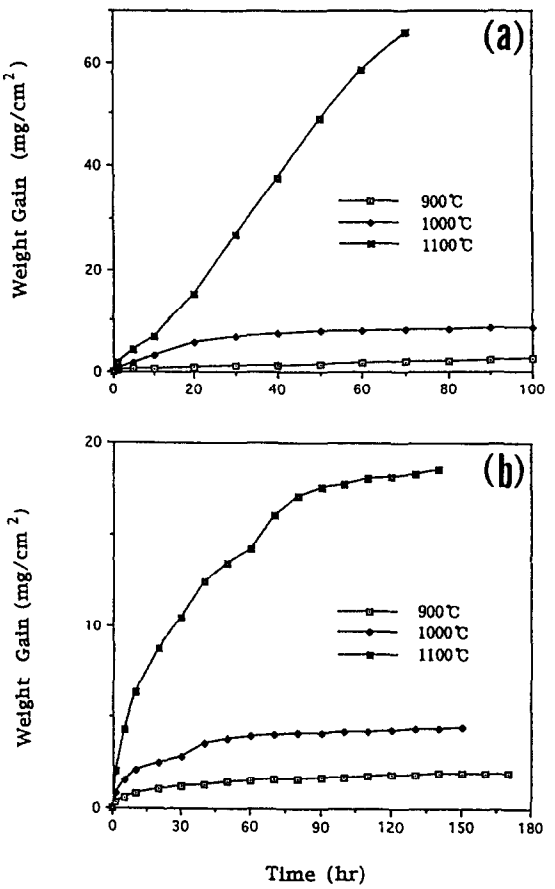


Figure 2. Weight change per unit area vs. oxidation time in air from 900~1100°C for (a) TiAl-5wt%Si, and (b) TiAl-5wt%Nb.

Experimental

The materials used in this study were TiAl, and TiAl intermetallic with addition of Cr, V, Si, Mo and Nb. The alloys were plasma-arc melted and casted in the form of cylindrical bars having a diameter of 12 mm. The chemical compositions of the TiAl alloys used in this experiment are shown in Table 1. The specimens used in the oxidation test were 10 mm in diameter and 2 mm thick. The specimens were polished with up to 600 grit SiC papers, and cleaned with acetone in an ultrasonic bath before oxidation.

Isothermal oxidation experiments were performed at 800~1100°C under the atmospheric environment in a vertical furnace. A Cahn-2000 microbalance in conjunction with a strip chart recorder was used to measure the weight of the specimen being oxidized. For marker experiments, specimens, wound by Pt-wire, were oxidized at 1000°C in air. Cyclic oxidations were carried out at 900 and 1000°C in air in a horizontal tube furnace. Each cycle consisted of 45 minutes at reaction temperature and 15 minutes at room temperature. The specimens were weighed after every 5 to 10 cycles at the beginning of the test and, later on, after every 24 cycles. Reaction products formed during the oxidation were examined by XRD and SEM equipped with WDX.

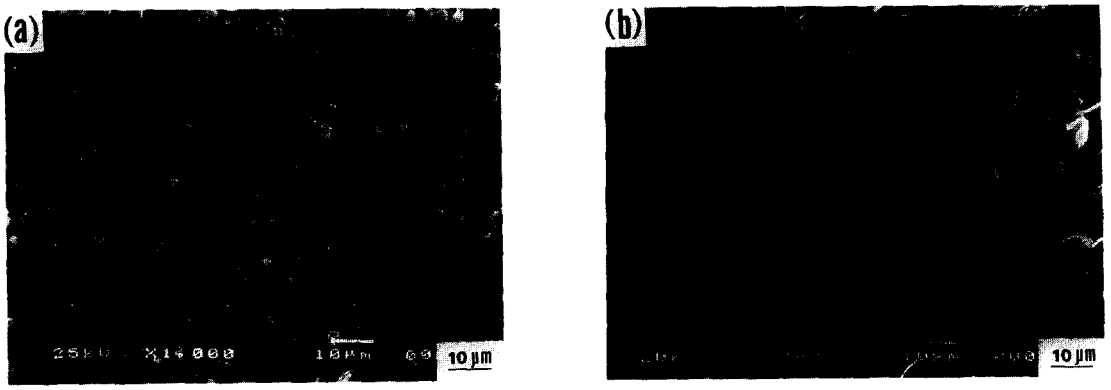


Figure 3. SEM photographs of surface oxides formed on TiAl-5wt%Nb oxidized in air (a) at 900°C for 170 hrs, and (b) at 1100°C for 140 hrs.

Results and Discussion

Fig. 1 shows the variation of weight gain per unit area of original specimen with oxidation time at 900°C. In the early stage of oxidation, the weight gains showed a parabolic shape, indicating formation of a thin, protective oxide scale(Al_2O_3). After the initial period, the weight gains became linear. The slope of the curves increased with the addition of Cr and V. In particular, the weight gain of TiAl-5wt%Cr alloy was the greatest among the alloys tested, and the weight gain curve of TiAl-5wt%V alloy showed discontinuity, probably due to cracking or spallation of oxide scale. However, the rate of oxidation decreased with increasing Cr content. Therefore, Cr and V seemed to be detrimental to oxidation resistance. The alloys with Si, Mo and Nb showed less weight gain than TiAl. The alloy with Nb showed much better oxidation resistance than the others. Weight gain curves of the alloys with Si and Nb were different in shape during the oxidation up to 40 hours, which was in good agreement with the study on oxidation of TiAl-1wt%Si and TiAl-1wt%Nb at 900°C by Maki, et al. (21) During the early stage of oxidation from 0 to 5 hours, the weight gain of TiAl-1wt%Si alloy was less than that of TiAl-1wt%Nb. And, they suggested that the mechanism for improving oxidation resistance by Si addition differed considerably from by Nb. The difference will be discussed later.

Fig. 2 shows the weight gain curves of TiAl-5wt%Si and TiAl-5wt%Nb alloys with oxidation time at various temperatures. The transient period from parabolic to linear part was shortened with increasing temperature. It was obvious that the oxidation resistance of TiAl-5wt%Nb alloy was better than that of TiAl-5wt%Si alloy. The tendency became remarkable at higher temperature. The oxidation rate of TiAl-5wt%Nb alloy was much slower than that of TiAl-5wt%Si alloy, roughly by a factor of 2 at 1000°C for 100 hours and 3 at 1100°C for 70 hours. However, weight gain curves of TiAl-5wt%Si and TiAl-5wt%Nb alloys were also different in shape at 1000 and 1100°C. Although the breakaway oxidation caused by spallation was observed for TiAl-5wt%Nb, the weight gain was not significant compared to that of TiAl-5wt%Si. However, the breakaway oxidation due to cracking in the oxide scale was observed at 1100°C for TiAl-5wt%Si. After the breakaway oxidation, the oxidation rate of TiAl-5wt%Si obeyed almost linear kinetics with rapid weight gain.

The morphology of surface oxide scales is critical in maintaining the oxide layer as a protective diffusion barrier. It was known that, during the very initial periods of TiAl oxidation at 1400°K for 1 hour in purified oxygen, Al_2O_3 and TiO_2 were formed on the surface where the upper right is covered with Al_2O_3 and TiO_2 crystals, while the lower left with TiO_2 crystals (13). The surface morphologies of oxide scale formed on TiAl-5wt%Nb alloy are shown in Fig. 3. The oxide formed on the alloys was finer and smoother at lower temperature as shown in Fig. 3. As the temperature increased, oxides formed on alloys became coarser. The crystal growth of randomly oriented rod-like grains of TiO_2 led to the formation of numerous voids and pores

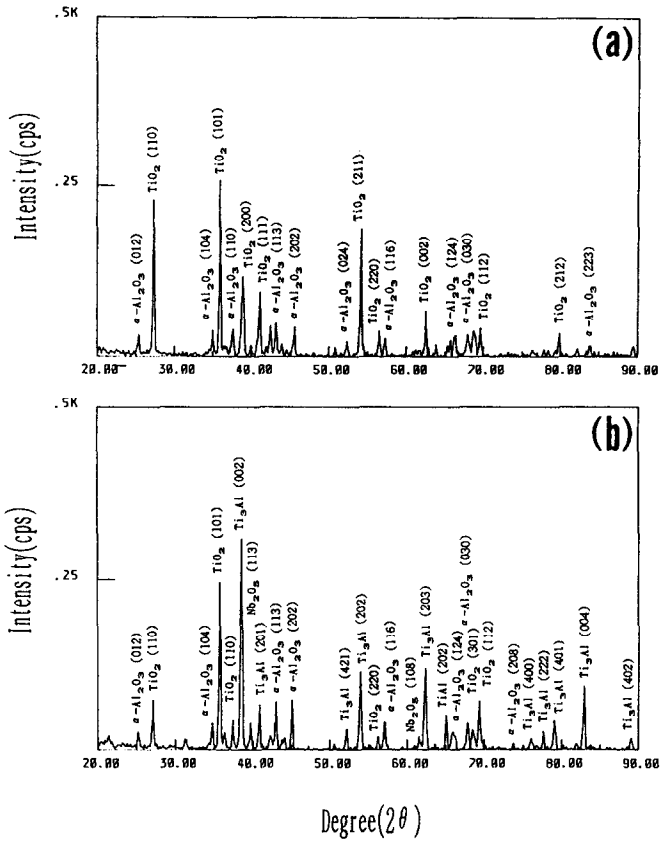


Figure 4. X-ray diffraction spectra of surface oxides formed in air at 900°C for (a) 100 hrs on TiAl-5wt%Si, and (b) 170 hrs. on TiAl-5wt%Nb.

at the grain boundaries which may initiate cracking in oxide layers and spallation of the oxides from the alloy substrates (7). Since TiO_2 crystals grew much faster than Al_2O_3, the outer surface was subsequently covered with a layer, mainly TiO_2 at higher temperature. And, cracks, which caused the breakaway oxidation as shown in Fig. 2(a), were observed on surface layer of TiAl-5wt%Si alloy oxidized at 1100°C.

Representative XRD patterns of oxidized TiAl-5wt%Si and TiAl-5wt%Nb specimens are shown in Fig. 4. In the TiAl-5wt%Si alloy, TiO_2(rutile) and α-Al_2O_3 were detected, but SiO_2 was not. Instead, TiAl and Ti_3Al peaks were observed in the TiAl-5wt%Nb alloy. Al_2O_3 peaks were relatively weak compared with TiO_2 and Ti_3Al peaks. It was believed that the Al_2O_3 layer formed on TiAl-5wt%Nb was thin, but protective. However, it could not be excluded that the content of Al was lower than the critical content of Al to form Al_2O_3 during oxidation, or the oxide scale produced was spalled during cooling. Ti_3Al was believed to be formed by the depletion of Al in TiAl. These patterns were also observed in thermal cyclic oxidation.

The cross sections of the oxide layer examined by SEM-WDX for TiAl-5wt%Si and TiAl-5wt%Nb alloys are shown in Figs. 5 and 6, respectively. For TiAl without Si and Nb addition, a continuous Al_2O_3 layer which acts as a barrier against oxidation was not formed. However, although the surface layer was mainly Al_2O_3 when oxidized at 900°C, TiO_2 was observed on the surface layer when oxidized at 1000 and 1100°C for TiAl-5wt%Si, as shown in Fig. 5. Al_2O_3 formed in the early oxidation stage was observed on the surface. The growth rate of TiO_2 is faster than that of Al_2O_3 with increasing temperature. Then, as a result of randomly oriented rod-like or lateral growth of TiO_2 crystals (7,13) with increasing temperature, cracks developed in

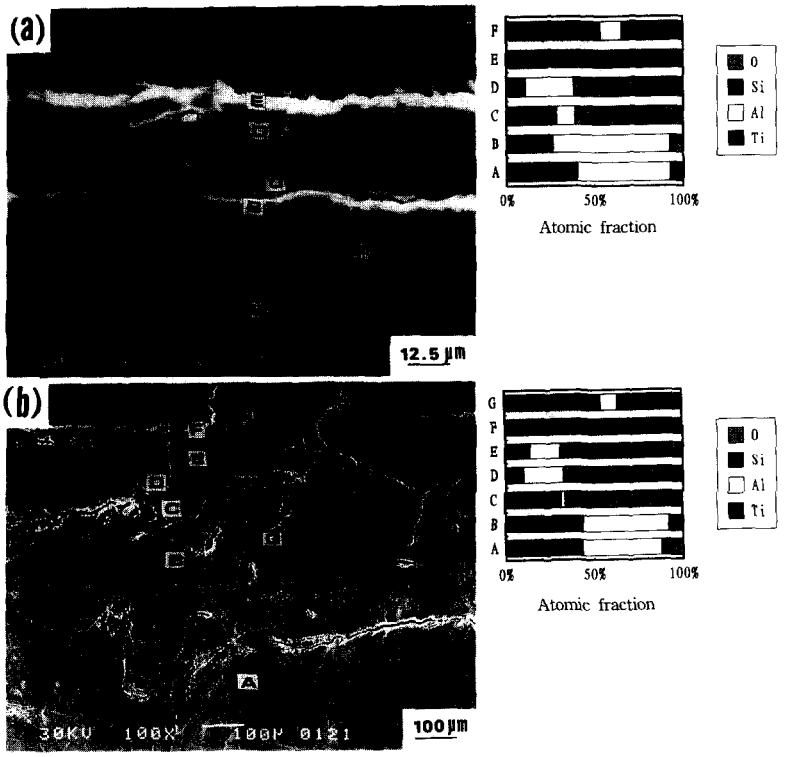


Figure 5. SEM photographs and WDX analyses of a cross-section of TiAl-5wt%Si oxidized in air (a) at 900°C for 100 hrs, and (b) at 1100°C for 70 hrs.

the surface layer and more internal oxidation occurred. More voids and pores were observed in the scale, and at the interface between oxide and substrate with increasing temperature. WDX analysis showed that Si was detected in the mixed layer of TiO_2+Al_2O_3, and seemed to form of SiO_2. Si formed crystalline and/or amorphous SiO_2 in the oxide layer (9,21). This SiO_2 can act as a barrier against oxidation, together with the Al_2O_3. TiSi compounds were detected in the matrix. The scales formed on TiAl-5wt%Nb alloy are shown in Fig. 6. The surface layer oxidized in air at 900°C for 170 hours was a mixture of $\underline{Al_2O_3}$ and TiO_2 (the underline indicates a major oxide), and the inner layer was composed of $\underline{TiO_2}/Al_2O_3$ and a small amount of Nb oxide. The most stable Nb oxide is Nb_2O_5 which can form compound oxide with each TiO_2 and Al_2O_3. For specimens oxidized in air at 1000°C for 150 hours and at 1100°C for 140 hours, the surface layer was considered to be a mixture of $\underline{TiO_2}/Al_2O_3$ because the amount of Al was not sufficient to form Al_2O_3. According to the study of Meier (22), the critical fraction of Al for forming a continuous Al_2O_3 for Ti-Al system was larger than about 46wt%(50at%). So, it was believed that the mixed layer of $\underline{Al_2O_3}/TiO_2$ with a very small amount of Nb acted as a relatively protective layer at 900°, but the mixture layer of $\underline{TiO_2}/Al_2O_3$ did not at 1000 and 1100°C. And, the internal oxidation in the matrix was observed with increasing temperature. Nb as an alloying element may decrease solubility and diffusion of oxygen or promote diffusion of Al (17). TiO_2+Nb_2O_5 may impede the inward diffusion of oxygen into substrate (18). More voids and pores were observed with increasing temperature. The tendency in thermal cyclic oxidation was similar to that in isothermal oxidation. It can be concluded that the greater oxidation resistance of the Nb-containing scale in air 900°C appears to be related to its ability to form relatively dense compound oxides rather than the mixture of TiO_2 and Al_2O_3.

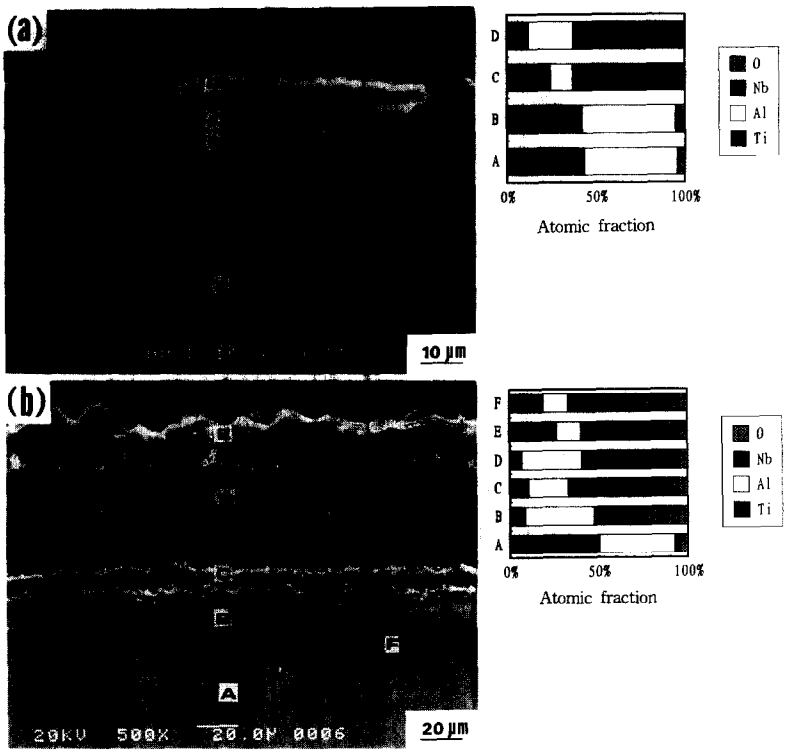


Figure 6. SEM photographs and WDX analyses of a cross-section of TiAl-5wt%Nb oxidized in air (a) at 900°C for 170 hrs, and (b) at 1100°C for 140 hrs.

Fig. 7 shows SEM photographs of cross-sections of specimens by Pt-marker experiments for TiAl, TiAl-5wt%Si and TiAl-5wt%Nb oxidized in air at 1000°C. The marker was located inside the outer scale(TiO_2) as shown in Fig. 7(a). It could be shown experimentally that in TiAl the diffusion process involved the simultaneous metal (mainly Ti ions) and oxygen diffusion through scale. On the other hand, according to the Pt-marker test of TiAl-5wt%Nb and TiAl-5wt%Si, oxygen diffused predominantly inward. But, in TiAl-5wt%Si it was considered that the location of the marker was influenced by plastic flow of the oxide scale around marker.

Weight-change data by the cyclic oxidation at 900°C in air is shown in Fig. 8. The alloys with Cr and Nb showed weight gain, which indicated less scale spallation than that of others. The spallation of the surface oxide scale could be enhanced by the thermal stress, resulting in the acceleration of oxidation. The addition of 5wt%Nb improved cyclic oxidation resistance at 900°C due to the improvement of scale adherence. However, weight change data of TiAl-5wt%Nb alloy by cyclic oxidation at 1000°C did not show any improvement of cyclic oxidation resistance.

Conclusions

1. The addition of Mo, Si and Nb with the order of 5wt% improved oxidation resistance of TiAl alloys for the isothermal oxidation at elevated temperature under atmospheric environment.
2. The adherence of oxide scale to the matrix was improved by the addition of Nb in TiAl intermetallic for thermal cyclic oxidation at 900°C in air.

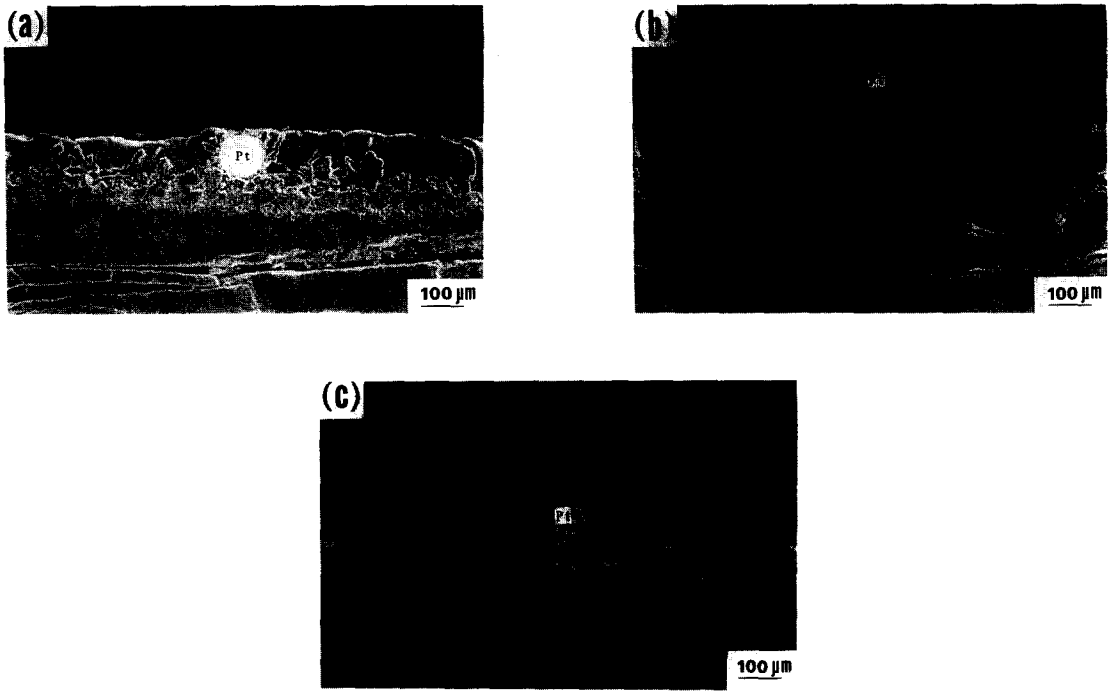


Figure 7. SEM photographs of cross-sections of specimens by Pt-marker experiments for (a) TiAl and (b) TiAl-5wt%Si oxidized in air at 1000°C for 150 hrs, and (c) TiAl-5wt%Nb oxidized in air at 1000°C for 200 hrs.

3. The effects of alloying elements, Si and Nb, on oxidation were quite different. Crystalline and/or amorphous SiO_2 in the oxide layer was formed by the addition of Si, which acted as a barrier against oxidation. Nb enhanced the formation of Al_2O_3, and changed the structure of the diffusion barrier against

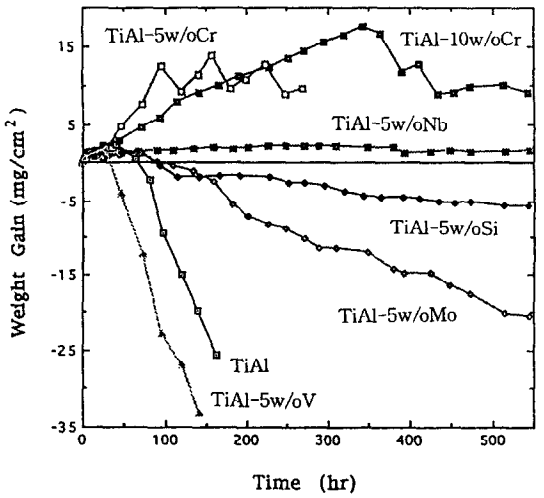


Figure 8. Weight change on thermal cyclic oxidation for specimens oxidized in air at 900°C.

oxidation. The $TiO_2/Al_2O_3/Nb_2O_5$ layer in the oxide scale was also protective while the TiO_2 was not considered to be protective.

4. According to the Pt-marker test, the oxidation mechanism of TiAl was by the simultaneous Ti-ions and oxygen diffusion, and that of TiAl-5wt%Si and TiAl-5wt%Nb was by the predominant inward diffusion of oxygen.

Acknowledgements

This work was supported by Korea Research Institute of Standards and Science.

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